

Rosen — §6.5: Generalized Permutations and Combinations

Selected Exercises

Alston

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Exercise 30

How many different strings can be made from the letters in MISSISSIPPI, using all the letters?

Exercise 40

How many ways are there to travel in $xyzw$ space from the origin $(0, 0, 0, 0)$ to the point $(4, 3, 5, 4)$ by taking steps one unit in the positive x , positive y , positive z , or positive w direction?

Exercise 46

A shelf holds 12 books in a row. How many ways are there to choose five books so that no two adjacent books are chosen? [*Hint*: Represent the books that are chosen by bars and the books not chosen by stars. Count the number of sequences of five bars and seven stars so that no two bars are adjacent.]

Exercise 48

Recall: Theorem 3 states that the number of different permutations of n objects, where there are n_1 indistinguishable objects of type 1, n_2 indistinguishable objects of type 2, $\dots$, and n_k indistinguishable objects of type k , is $n!/(n_1!n_2!\cdots n_k!)$. Theorem 4 states that the number of ways to distribute n distinguishable objects into k distinguishable boxes so that n_i objects are placed into box i , $i = 1, 2, \dots, k$, equals $n!/(n_1!n_2!\cdots n_k!)$.

Prove Theorem 4 by first setting up a one-to-one correspondence between permutations of n objects with n_i indistinguishable objects of type i , $i = 1, 2, 3, \dots, k$, and the distributions of n objects in k boxes such that n_i objects are placed in box i , $i = 1, 2, 3, \dots, k$, and then applying Theorem 3.

Exercise 49

Recall: Theorem 2 states that there are $\binom{n+r-1}{r} = \binom{n+r-1}{n-1}$ r -combinations from a set with n elements when repetition of elements is allowed.

In this exercise we will prove Theorem 2 by setting up a one-to-one correspondence between the set of r -combinations with repetition allowed of $S = \{1, 2, 3, \dots, n\}$ and the set of r -combinations of the set $T = \{1, 2, 3, \dots, n + r - 1\}$.

- Arrange the elements in an r -combination, with repetition allowed, of S into an increasing sequence $x_1 \leq x_2 \leq \cdots \leq x_r$. Show that the sequence formed by adding $k - 1$ to the k th term is strictly increasing. Conclude that this sequence is made up of r distinct elements from T .
- Show that the procedure described in (a) defines a one-to-one correspondence between the set of r -combinations, with repetition allowed, of S and the r -combinations of T . [*Hint*: Show

the correspondence can be reversed by associating to the r -combination $\{x_1, x_2, \dots, x_r\}$ of T , with $1 \leq x_1 < x_2 < \dots < x_r \leq n + r - 1$, the r -combination with repetition allowed from S , formed by subtracting $k - 1$ from the k th element.]

- (c) Conclude that there are $\binom{n+r-1}{r}$ r -combinations with repetition allowed from a set with n elements.

Exercise 59

How many ways are there to distribute five balls into three boxes if each box must have at least one ball in it if

- (a) both the balls and boxes are labeled?
- (b) the balls are labeled, but the boxes are unlabeled?
- (c) the balls are unlabeled, but the boxes are labeled?
- (d) both the balls and boxes are unlabeled?

Exercise 61

Suppose that a weapons inspector must inspect each of five different sites twice, visiting one site per day. The inspector is free to select the order in which to visit these sites, but cannot visit site X , the most suspicious site, on two consecutive days. In how many different orders can the inspector visit these sites?

Exercise 63

Prove the Multinomial Theorem: If n is a positive integer, then

$$(x_1 + x_2 + \dots + x_m)^n = \sum_{n_1 + n_2 + \dots + n_m = n} C(n; n_1, n_2, \dots, n_m) x_1^{n_1} x_2^{n_2} \dots x_m^{n_m},$$

where

$$C(n; n_1, n_2, \dots, n_m) = \frac{n!}{n_1! n_2! \dots n_m!}$$

is a multinomial coefficient.